



Brief paper

Optimal deployment of vehicles with circular formation for bearings-only multi-target localization[☆]Yinya Li^{*}, Guoqing Qi, Andong Sheng

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ABSTRACT

This paper focuses on the optimal deployment problem of multiple mobile vehicles with circular formation for multi-target localization using bearings-only sensors. Initially, a general closed-form formula of geometric dilution of precision (GDOP) is derived based on the Cramer–Rao lower bound (CRLB) for bearings-only localization. By this generalized formula, the analytical formula of GDOP for multiple vehicles containing the same observation noise variance with circular formation is then deduced directly. Next, the necessary and sufficient condition for sensor placement optimality with the minimization of GDOP is presented, which guarantees what are the best positions where the vehicles should be in the desired observation circumference with the given radius centered at the target. The optimal deployment positions as well as the global minimum of GDOP are then proposed with analytical expressions. Furthermore, an efficient optimal deployment algorithm is developed to deploy the vehicles to approach the best positions as precisely as possible for practical multi-target localization. Finally, an illustrative example is used to demonstrate the effectiveness of the proposed approach.

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1. Introduction

Bearings-only target localization is one of the most used ways of source localization (Farina, 1999; Nardone & Aidala, 1981), which can be traced back to the submarine/ship localization in underwater information counterworks (Rao, Murthy, & Rajeswari, 2010; Tan, Diamant, Seah, & Waldmeyer, 2011). Nowadays, bearings-only sensors as well as other sensors, such as range-only sensors, time-difference-of-arrival sensors and frequency-difference-of-arrival sensors, are extensively used in wireless sensor networks (Hmam, 2017; Kim, Kim, Han, Song, & Kim, 2016; Meng, Xiao and Xie, 2013; Meng, Xie and Xiao, 2013; Younis & Akkaya, 2008). Compared with other sensors, bearings-only sensors operate secretly in passive mode and are not easy to be attacked by electronic jammers and anti-radiation missiles. Thus, they have the advantage of survival ability in the environment of modern electronic warfare and play a key role

in a number of surveillance systems for defense (Farina, 1999). Moreover, there usually exist the scenarios in practical surveillance systems where more than one sensor is needed to achieve cooperative localization with some special geometric formations. Among which, circular formation is the most popularly used one and can be applied in the scenarios where sensors are required to enclose, capture, secure or monitor a target (Yu & Liu, 2016, 2017). In this paper, we will investigate the optimal deployment problem of vehicles with circular formation for multi-target localization using bearings-only sensors.

It is well known that the sensor-target geometry has a significant impact on the performance of target localization (Bar-Shalom, Li, & Kirubarajan, 2004). The popularly used criteria for assessing the accuracy of target localization are the Fisher information matrix (FIM) and its inverse (Doğançay & Hmam, 2008). The inverse of the FIM (when it exists) is called the Cramer–Rao lower bound (CRLB). The corresponding optimal sensor placement can be categorized as two types: A-optimality (Doğançay & Hmam, 2008) and D-optimality (Bishop, Fidan, Anderson, Doğançay, & Pathirana, 2010; Xu & Doğançay, 2017). A-optimality is dedicated to minimizing the trace of the inverse of the FIM, while D-optimality seeks to maximize the determinant of the FIM. The geometric dilution of precision (GDOP) is defined as the trace of the CRLB, and is also used to the optimal sensor placement (Chaffee & Abel, 1994; Levanon, 2000; Sharp, Yu, & Guo, 2009). The GDOP shows how the localization accuracy is

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to be affected by the sensor-target geometry, which can be used to evaluate the final localization estimation (Meng, Xie, & Xiao, 2016).

The optimal sensor placement design based on A-optimality and/or D-optimality for single target localization has been extensively studied in Bishop et al. (2010), Doğançay and Hmam (2008), MartíNez and Bullo (2006), Rensfelt, Mousavi, Mossberg, and Soderstrom (2008) and Xu and Doğançay (2017), to name a few. One of the remarkable works on optimal sensor placement is the one developed by MartíNez and Bullo (2006), in which the optimal sensor-target geometries are obtained based on D-optimality using range-only sensors. Recently, a comprehensive set of studies for optimal sensor placement has been proposed, see Bishop et al. (2010), Jiang, Bishop, Anderson, and Drake (2015), Perelman, Abbas, Koutsoukos, and Amin (2016), Xu and Doğançay (2017) and Zhao, Chen, and Lee (2013). Particularly, in Zhao et al. (2013), the frame theory is first applied to the optimal sensor placement for single target localization, and which can be regarded as the generalization of the existing works on optimal sensor placement.

Until now, most of the existing studies only focus on finding the necessary conditions for sensor placement optimality (Bishop et al., 2010; Doğançay & Hmam, 2008; Xu & Doğançay, 2017), and the sufficient ones for optimal sensor placement have been less considered. Moreover, most of the existing works only consider the optimal sensor placement for single target localization (Bishop et al., 2010; Doğançay & Hmam, 2008; MartíNez & Bullo, 2006; Xu & Doğançay, 2017; Zhao et al., 2013). Very few studies in the literature have considered the problem of optimal sensor placement for multi-target localization. However, in practical applications, there usually exist the scenarios where more than one target needs to be localized simultaneously by n sensors. Particularly, for some tactical purpose, n sensors are further required to be decomposed into several groups to realize the optimality of sensor placement for each target localization simultaneously. Naturally, two closely related problems can arise in the optimal sensor placement for multi-target localization. The first is to determine how many sensors should be used for each target, and the second is how to group them together so that they are deployed with the best positions to fulfill specific target localization tasks. Actually, these two problems are inherently included in the types of linear and nonlinear integer programming (LIP and NLIP) ones (Floudas, 1995; Li & Sun, 2006). Therefore, some optimization techniques should be adopted to handle these problems for multi-target localization.

Motivated by the aforementioned aspects, this paper focuses on the optimal deployment problem of vehicles with circular formation for multi-target localization using bearings-only sensors. In this case, n mobile vehicles are first distributed randomly in the region of targets, then the vehicles are redeployed and forced to be gathered around the assigned target with circular formation to estimate the target location. This is because in surveillance systems the targets to be localized may represent some threats (Dasgupta, Ibeawuchi, & Ding, 2010), or for some tactical purpose, the vehicles should not be exposed (Moghe, Consolazio, & Fudem, 1995; Ronald Schineller, 1995). Thus, the constraint of no vehicle be closer than a certain distance from the assigned target with circular formation is mainly considered in this paper.

The main contribution of this paper is threefold: First, a general closed-form formula of GDOP for $n \geq 2$ bearings-only sensors with arbitrary observation distances and noise variances is derived. Second, a necessary and sufficient condition for sensor placement optimality with circular formation is presented. All the optimal sensor-target geometries are directly obtained by the analytical solutions of optimal sensor placement angles. Third,

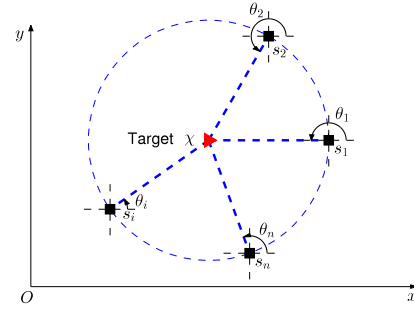


Fig. 1. Illustration of sensor-target geometry with circular formation using bearings-only measurements.

an efficient optimal deployment algorithm of vehicles for practical multi-target localization is developed. This algorithm can be further inherently integrated into the circular formation control strategies, as the proposed approaches in Yu and Liu (2016) and Yu and Liu (2017).

This paper is organized as follows: Section 2 presents the preliminaries including the problem formulation and two technical lemmas. The main results are given in Section 3. In Section 4, an illustrative example is proposed, and Section 5 concludes the paper and outlines areas for future research.

Notations: The 2-norm for a vector of $\mathbf{x} \in \mathbb{R}^n$ is defined as $\|\mathbf{x}\| = \sqrt{\mathbf{x}^T \mathbf{x}}$, $\text{tr}\{\cdot\}$ and $\det(\cdot)$ denote the trace and the determinant of the matrix contained in the bracket, respectively.

2. Preliminaries

2.1. Problem formulation

Consider the localization problem of n ($n \geq 2$) mobile vehicles around a ground stationary target with circular formation using bearings-only measurements, shown in Fig. 1, where $\chi := [x_t, y_t]^T \in \mathbb{R}^2$ and $\mathbf{s}_i := [x_i, y_i]^T \in \mathbb{R}^2$ are the mutually distinct position vectors of the target and vehicles, respectively, $i \in \{1, 2, \dots, n\}$. θ_i is the angle of line-of-sight (LOS) from vehicle i to the target, given by

$$\theta_i = h_i(\chi) = \tan^{-1} \left(\frac{y_t - y_i}{x_t - x_i} \right) \quad (1)$$

where $\tan^{-1}(\cdot)$ is the four-quadrant tangent function. Define z_i as the observation of θ_i , then the observation equation is

$$z_i = \theta_i + v_i = h_i(\chi) + v_i \quad (2)$$

where v_i is the observation error, and assumed to be i.i.d. Gaussian noise with mean zero and variance $\sigma_{\theta_i}^2$, i.e., $v_i \sim \mathcal{N}(0, \sigma_{\theta_i}^2)$, $i \in \{1, 2, \dots, n\}$.

In practice, the target position can be estimated by using the bearings-only observations of z_i . Denote $\hat{\chi}$ as the estimate of χ , then the estimation error covariance matrix can be defined as $\mathbf{P} = \mathcal{E}\{(\chi - \hat{\chi})(\chi - \hat{\chi})^T\}$. It is well known that, the inverse of the FIM (when it exists, called the CRLB), is the lower bound of the covariance matrix $\mathbf{P}$ for any unbiased estimator, satisfying the following matrix inequality (Bar-Shalom et al., 2004)

$$\mathbf{P} \geq \mathbf{J}^{-1} = \text{CRLB} \quad (3)$$

where $\mathbf{J}$ is the FIM. Then the GDOP is defined as

$$\text{GDOP} = \sqrt{\text{tr}\{\text{CRLB}\}} \quad (4)$$

In this paper, the optimal sensor placement design is based on the minimization of the GDOP in (4). For multi-target localization, the

considered optimal deployment problem of vehicles with circular formation can be decomposed into the following two decision problems:

- [Q1] How many vehicles should be used for each target?
 [Q2] For each target, how to group them together so that they are placed with the best positions in the desired observation circumference to achieve target localization?

For solving the decision problems [Q1]–[Q2], the following three closely related problems are formally defined, where the solution to [Problem 2](#) depends on [Problem 1](#), and the solution to [Problem 3](#) depends on [Problem 2](#). [Problem 3](#) is used to solve the decision problem [Q1], whereas [Problem 2](#) is defined for solving the decision problem [Q2].

Problem 1. Define the distance from vehicle i to the target as $d_i = \|\mathbf{x} - \mathbf{s}_i\|$. For bearings-only target localization, determine the explicit formula of the GDOP in (4) with the form of

$$\text{GDOP} = \psi(\boldsymbol{\theta}, \mathbf{d}, \boldsymbol{\sigma}, n) \quad (5)$$

where $\boldsymbol{\theta} = [\theta_1, \theta_2, \dots, \theta_n]^T$, $\mathbf{d} = [d_1, d_2, \dots, d_n]^T$ and $\boldsymbol{\sigma} = [\sigma_{\theta_1}, \sigma_{\theta_2}, \dots, \sigma_{\theta_n}]^T$.

Problem 2. Denote r as the given radius of the desired observation circumference and define $\Delta\theta_l = \theta_{l+1} - \theta_l$, $l \in \{1, 2, \dots, n-1\}$. For $\Delta\theta_1 = \Delta\theta_2 = \dots = \Delta\theta_{n-1}$, $d_i = r$, $\sigma_{\theta_i} = \sigma_\theta$, $\forall i \in \{1, 2, \dots, n\}$, find all the optimal deployment solutions

$$\boldsymbol{\theta}^* = [\theta_1^*, \theta_2^*, \dots, \theta_n^*]^T \quad (6)$$

for this desired (ideal) circular formation such that the GDOP in (5) reaches its global minimum

$$\text{GDOP}_{\min} = \varphi(r, \sigma_\theta, n) \quad (7)$$

Problem 3. Given the observation radiuses of r_ℓ , $\ell \in \{1, 2, \dots, U\}$ for $U \geq 2$ targets and $n \geq 2U$ vehicles with the same noise variance of σ_θ^2 . Determine the allocating number n_ℓ of vehicles for each target, such that the sum of squared errors of GDOP

$$\kappa = \sum_{1 \leq p < q \leq U} [\text{GDOP}_{\min}^p - \text{GDOP}_{\min}^q]^2 \quad (8)$$

is minimized with the constraints of $\sum_{\ell=1}^U n_\ell = n$ and $n_\ell \geq 2$, where $\text{GDOP}_{\min}^p = \varphi(r_p, \sigma_\theta, n_p)$ and $\text{GDOP}_{\min}^q = \varphi(r_q, \sigma_\theta, n_q)$ denote the global minimum determined by (7) for target p and target q , respectively.

2.2. Two technical lemmas

Here, two technical lemmas ([Lemmas 4](#) and [5](#)) are first given, which will be used to prove the minimization of GDOP for [Problem 2](#), and the corresponding proofs are presented in [Appendices A](#) and [B](#), respectively.

Consider the function

$$f(x) = \sum_{i=1}^{n-1} (n-i) \sin^2(ix) \quad (9)$$

where $i \in \{1, 2, \dots, n-1\}$, $x \in \mathbb{R}$.

Lemma 4. For all $x \in \mathbb{R}$, the closed forms of $f(x)$, its first-order derivative of $df(x)/dx$ and its second-order derivative of $d^2f(x)/dx^2$, respectively, are

$$f(x) = \begin{cases} 0, & x = k\pi \\ \frac{n^2 \sin^2(x) - \sin^2(nx)}{4 \sin^2(x)}, & \text{otherwise} \end{cases} \quad (10)$$

$$\frac{df(x)}{dx} = \begin{cases} 0, & x = k\pi \\ \frac{\alpha(x)}{2 \sin^3(x)}, & \text{otherwise} \end{cases} \quad (11)$$

$$\frac{d^2f(x)}{dx^2} = \begin{cases} n^2(n^2 - 1)/6, & x = k\pi \\ \frac{\beta(x)}{2 \sin^4(x)}, & \text{otherwise} \end{cases} \quad (12)$$

where $k \in \mathbb{Z}$ and

$$\alpha(x) = \sin(nx)[\sin(nx) \cos(x) - n \cos(nx) \sin(x)] \quad (13)$$

$$\beta(x) = -n^2 \sin^2(x) \cos(2nx) + n \sin(2x) \sin(2nx) - [\cos(2x) + 2] \sin^2(nx) \quad (14)$$

Lemma 5. The function of $f(x)$ in (9) reaches its global maximum

$$f_{\max} = n^2/4 \quad (15)$$

if and only if

$$x^* = k\pi/n \quad (16)$$

where $k \in \Phi$ and $\Phi = \{k \mid k \neq mn, k \in \mathbb{Z}, m \in \mathbb{Z}\}$.

3. Main results

3.1. Explicit formula of GDOP

We first give the theorem to derive the explicit formula of the GDOP in (5) for [Problem 1](#) as below.

Theorem 6. Consider the bearings-only localization system with $n \geq 2$ vehicles around the target as shown in [Fig. 1](#). The explicit formula of the GDOP in (5) is determined as

$$\text{GDOP} = \left(\frac{\sum_{i=1}^n c_i^2}{\sum_{1 \leq i < j \leq n} c_i^2 c_j^2 \sin^2(\theta_{ij})} \right)^{1/2} \quad (17)$$

where $\theta_{ij} = \theta_j - \theta_i$ and $c_i = 1/(d_i \sigma_{\theta_i})$, $i \in \{1, 2, \dots, n\}$.

Proof. See [Appendix C](#).

Remark 7. Note that any additional restrictive assumption is not made on the number of sensors, the observation ranges and noise variances for the derivation of the GDOP in (17). The formulas of the determinant $\det(\mathbf{J})$ with identical observation ranges and/or identical noise variances can be found in [Bishop et al. \(2010\)](#), [Doğançay and Hmam \(2008\)](#) and [Farina \(1999\)](#), which can be viewed as the special cases of (C.7) and can be immediately obtained just by setting $d_i = r$ and/or $\sigma_{\theta_i} = \sigma_\theta$, $\forall i \in \{1, 2, \dots, n\}$.

As noted in [Problem 2](#), in the desired (ideal) circular formation, $d_i = r$, $\sigma_{\theta_i} = \sigma_\theta$, $\forall i \in \{1, 2, \dots, n\}$, which shows that n vehicles are all required to be located at a circumference with a given observation radius r around the target. For this ideal circular formation, using (17), the GDOP can be readily computed as

$$\text{GDOP} = r \sigma_\theta \sqrt{\frac{n}{\sum_{1 \leq i < j \leq n} \sin^2(\theta_{ij})}} \quad (18)$$

In the latter sections, by minimizing the GDOP in (18), we will first show what are the best positions where the vehicles should be for this ideal circular formation. Then from a practical application perspective, we will present an efficient algorithm to deploy these vehicles to approach the best positions as precisely as possible.

3.2. Solutions of optimal deployment

Now the optimal deployment solutions on [Problem 2](#) are considered with the following theorem.

Theorem 8. Consider the minimization of GDOP in bearings-only localization systems with $n \geq 2$ vehicles around the target by circular formation. Given a desired observation radius r , for $d_i = r$, $\sigma_{\theta_i} = \sigma_\theta$, $\forall i \in \{1, 2, \dots, n\}$ and $\Delta\theta_1 = \Delta\theta_2 = \dots = \Delta\theta_{n-1}$, the GDOP in (18) reaches its global minimum of

$$\text{GDOP}_{\min} = \varphi(r, \sigma_\theta, n) = \frac{2r\sigma_\theta}{\sqrt{n}} \quad (19)$$

if and only if

$$\theta_i^* = \theta_1 + (i-1)\frac{k\pi}{n} \quad (20)$$

where $\theta_1 \in \mathbb{R}$, $k \in \Phi$ and $\Phi = \{k \mid k \neq mn, k \in \mathbb{Z}, m \in \mathbb{Z}\}$.

Proof. For this desired circular formation, it follows immediately that the minimization of the GDOP in (18) is equivalent to the maximization of

$$f(\theta) \triangleq \sum_{1 \leq i < j \leq n} \sin^2(\theta_{ij}) \quad (21)$$

Recall $\Delta\theta_l = \theta_{l+1} - \theta_l$, $l \in \{1, 2, \dots, n-1\}$, then we have

$$\theta_i = \begin{cases} \theta_1, & i = 1 \\ \theta_1 + \sum_{l=1}^{i-1} \Delta\theta_l, & i > 1 \end{cases} \quad (22)$$

Using (22), one obtains $\theta_{ij} = \sum_{l=i}^{j-1} \Delta\theta_l$. Since $\Delta\theta_1 = \Delta\theta_2 = \dots = \Delta\theta_{n-1}$, define $x = \Delta\theta_l$, $\forall l \in \{1, 2, \dots, n-1\}$, then $\theta_{ij} = \sum_{l=i}^{j-1} \Delta\theta_l = (j-i)x$. Submitting it into (21), the function $f(\theta)$ is then transformed into the function of $f(x)$ in (9). Thus, the GDOP in (18) can be rewritten as

$$\text{GDOP} = r\sigma_\theta \sqrt{\frac{n}{f(x)}} \quad (23)$$

where $x \neq k\pi$, $k \in \mathbb{Z}$. Note that $r > 0$, $\sigma_\theta > 0$ and $x \neq k\pi$, $k \in \mathbb{Z}$, $f(x) > 0$. From (23), it is readily observed that the GDOP reaches its global minimum as $f(x)$ reaches its global maximum, i.e.

$$\text{GDOP}_{\min} = \varphi(r, \sigma_\theta, n) = r\sigma_\theta \sqrt{\frac{n}{f_{\max}}} = \frac{2r\sigma_\theta}{\sqrt{n}} \quad (24)$$

Using [Lemma 5](#), it is shown that the function of $f(x)$ reaches its global maximum $f_{\max} = n^2/4$ if and only if $x = x^*$ as defined in (16). Noting that $x^* = \Delta\theta_l^*$ and using (16) in (22), it follows immediately that the GDOP in (18) reaches its global minimum in (24) if and only if $\theta_i^* = \theta_1 + (i-1)k\pi/n$, $\theta_1 \in \mathbb{R}$, $i \in \{1, 2, \dots, n\}$ and $k \in \Phi$, $\Phi = \{k \mid k \neq mn, k \in \mathbb{Z}, m \in \mathbb{Z}\}$. This completes the proof. ■

Remark 9. [Theorem 8](#) shows that the global optimal solutions of (20) are not only necessary but also sufficient. In such circular formation, the global optimal solutions proposed by [Bishop et al. \(2010\)](#) and [Doğançay and Hmam \(2008\)](#) are $2\pi/n$ (equiangular separations, $n \geq 3$) and π/n ($n \geq 2$). For $n \leq 3$, the optimal solutions of [Bishop et al. \(2010\)](#) and [Doğançay and Hmam \(2008\)](#) are the same as the ones defined by (20). However, for $n \geq 4$, the optimal solutions proposed by [Bishop et al. \(2010\)](#) and [Doğançay and Hmam \(2008\)](#) are the subset of our optimal solutions described by (20).

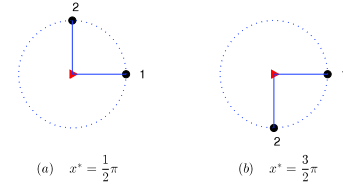


Fig. 2. Optimal deployment patterns for $n = 2$.

Remark 10. Note that the parameter of $\theta_1 \in \mathbb{R}$ in (20) is the angle of offset for the optimal deployment of vehicles and can be chosen arbitrarily. In practice, the angles of θ_i , $i \in \{1, 2, \dots, n\}$, are usually defined over the interval of $[0, 2\pi)$. Thus, the set of Φ can be compressed to the set of Φ_1 as below

$$\Phi_1 = \{k \mid 0 < k < 2n, k \neq n, k \in \mathbb{Z}\} \quad (25)$$

Then the number of the optimal solutions of θ_i^* in (20) for each vehicle (except the vehicle deployed by θ_1^*) becomes $2(n-1)$.

From [Theorem 8](#), the CRLB with the optimal solutions of θ_i^* in (20) can be analytically calculated. Define the elements of the main diagonal of CRLB in (C.5) as $[\sigma_{x_t}^{\text{CRLB}}]^2$ and $[\sigma_{y_t}^{\text{CRLB}}]^2$, respectively. Noting that $d_i = r$, $\sigma_{\theta_i} = \sigma_\theta$, $\forall i \in \{1, 2, \dots, n\}$ and according to (C.5), we determine that

$$[\sigma_{x_t}^{\text{CRLB}}(\theta)]^2 = r^2 \sigma_\theta^2 \frac{\sum_{i=1}^n \cos^2(\theta_i)}{\sum_{1 \leq i < j \leq n} \sin^2(\theta_{ij})} \quad (26)$$

$$[\sigma_{y_t}^{\text{CRLB}}(\theta)]^2 = r^2 \sigma_\theta^2 \frac{\sum_{i=1}^n \sin^2(\theta_i)}{\sum_{1 \leq i < j \leq n} \sin^2(\theta_{ij})} \quad (27)$$

Submitting θ_i^* into (26) and (27), respectively, yields

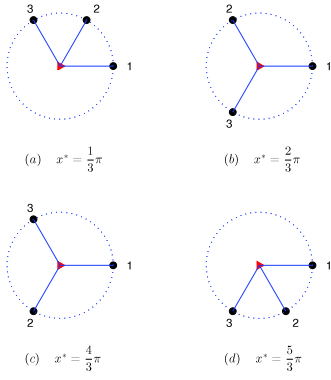
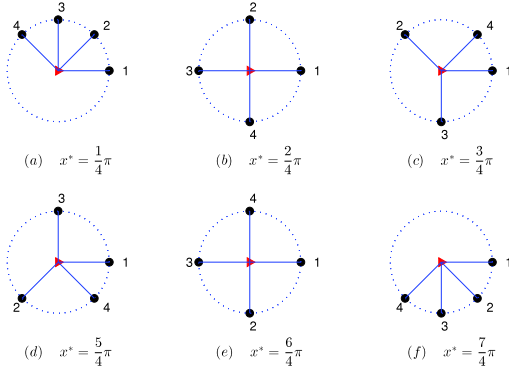
$$[\sigma_{x_t}^{\text{CRLB}}(\theta^*)]^2 = [\sigma_{y_t}^{\text{CRLB}}(\theta^*)]^2 = \frac{2r^2 \sigma_\theta^2}{n} \quad (28)$$

[Figs. 2–4](#) illustrate the optimal deployment patterns of vehicles for $n = 2, 3, 4$ with $\theta_1^* = \theta_1 = \pi$, respectively, where the red solid triangle represents the target, the black dots denote the optimal deployment points, and the numbers nearby the black dots are the labels of the corresponding optimal deployment points. From [Figs. 2–4](#), it is shown that the optimal deployment patterns are symmetric with the line governed by θ_1^* . For example, see [Fig. 4](#), (a) and (f), (b) and (e), (c) and (d) are symmetric with the line with respect to $\theta_1^* = \pi$, respectively. Meanwhile, it is observed from (18) that the global minimum of GDOP in (19) is independent of the following transformations: exchanging the labels of the optimal deployment points with arbitrary order, or flipping any optimal deployment point about the target, or any global rotation by choosing arbitrarily the angle of θ_1 , which coincides with the results of equivalent placements in [Zhao et al. \(2013\)](#).

3.3. Optimal allocating numbers of vehicles

In this section, the optimal allocating number of vehicles for each target on [Problem 3](#) is considered. First, using [Theorem 8](#), we transform the optimal allocating problem defined in [Problem 3](#) into an NLIP one, described by

$$\begin{aligned} \min \quad & \kappa = \sum_{1 \leq p < q \leq U} [\varphi(r_p, \sigma_\theta, n_p) - \varphi(r_q, \sigma_\theta, n_q)]^2 \\ \text{s.t.} \quad & \sum_{\ell=1}^U n_\ell = n \\ & n_\ell \in [2, n], n_\ell \in \mathbb{Z} \end{aligned} \quad (29)$$

Fig. 3. Optimal deployment patterns for $n = 3$.Fig. 4. Optimal deployment patterns for $n = 4$.

where κ is the error cost function to be minimized, and

$$\varphi(r_p, \sigma_\theta, n_p) = \frac{2r_p\sigma_\theta}{\sqrt{n_p}}, \quad \varphi(r_q, \sigma_\theta, n_q) = \frac{2r_q\sigma_\theta}{\sqrt{n_q}} \quad (30)$$

in which $r_p(r_q)$ and $n_p(n_q)$ are the given observation radius and the allocating number of vehicles for the p th(q th) target respectively.

Then, considering the symmetry of $\varphi(r_p, \sigma_\theta, n_p)$ and $\varphi(r_q, \sigma_\theta, n_q)$, the NLIP in (29) can be equivalently simplified as the following one, given by

$$\begin{aligned} \min \quad & \kappa = \sum_{p=1}^U [\varphi(r_p, \sigma_\theta, n_p) - c_0]^2 \\ \text{s.t.} \quad & \sum_{p=1}^U n_p = n \\ & n_p \in [2, n], \quad n_p \in \mathbb{Z} \end{aligned} \quad (31)$$

where

$$c_0 = \frac{2r_p\sigma_\theta}{\sqrt{n_p^*}}, \quad n_p^* = \frac{(r_p)^2}{\sum_{p=1}^U (r_p)^2} n \quad (32)$$

Finally, solving the NLIP in (31), the optimal allocating number n_p of vehicles for each target can be obtained.

Remark 11. The NLIP in (31) can be solved by a branch and bound method of Li and Sun (2006) and the optimal solution can then be found. However, the branch and bound methods for solving the NLIP problems usually have exponential time-complexity (Floudas, 1995; Li & Sun, 2006). For large scale multi-target localization systems or extending the approach for multi-target tracking, the greedy algorithms in Cormen, Leiserson, and Rivest (1990) can be used to solve the NLIP in (31) for decreasing the computational complexity.

3.4. Algorithm of optimal deployment

Now in this section, we are ready to develop an algorithm to deploy n mobile vehicles to approach the best positions as precisely as possible for multi-target localization.

In practice, the true target position is unknown to vehicles. However, the target position can be estimated with bearings-only measurements by the triangulation techniques (Bar-Shalom et al., 2004; Doğançay, 2004). Using the observation of z_i instead of θ_i in (1), then the set of the equations of LOS of n vehicles can be written as a compact form of

$$\mathbf{A}_i \boldsymbol{\chi} = b_i \quad (33)$$

where $\mathbf{A}_i = [\sin(z_i) \ -\cos(z_i)]$, $b_i = x_i \sin(z_i) - y_i \cos(z_i)$, $i \in \{1, 2, \dots, n\}$. Taking the least squares (LS) solution of (33) as the estimate of $\boldsymbol{\chi}$, then we obtain the closed form

$$\hat{\boldsymbol{\chi}} = [\hat{x}_t, \hat{y}_t]^T = \mathbf{A}^{-1} \mathbf{B} \quad (34)$$

where $\mathbf{A} = \sum_{i=1}^n \mathbf{A}_i^T \mathbf{A}_i$ and $\mathbf{B} = \sum_{i=1}^n \mathbf{A}_i^T b_i$.

Remark 12. It is worthwhile to mention that the estimator of LS in (34) is biased. It was shown in Doğançay (2004) that the bias of the estimator of LS in (34) is dependent on $(\bar{\mathbf{s}} - \boldsymbol{\chi})$ where $\bar{\mathbf{s}}$ is the mean of vehicles location, which means that a sensor-target geometry with small $\|\bar{\mathbf{s}} - \boldsymbol{\chi}\|$ will have a small bias. Therefore, it is recommended to choose the parameter of k with the smallest $\|\bar{\mathbf{s}} - \boldsymbol{\chi}\|$ if the estimator of LS in (34) is used.

Remark 13. An alternative method of determining the estimate of $\boldsymbol{\chi}$ is the maximum likelihood estimation (MLE), i.e.

$$\hat{\boldsymbol{\chi}} = \arg \max_{\boldsymbol{\chi}} \{\ln p(\mathbf{z}|\boldsymbol{\chi})\} \quad (35)$$

where $\ln p(\mathbf{z}|\boldsymbol{\chi})$ is the logarithmic likelihood function. However, the MLE has generally no closed-form solution to (35) (Bar-Shalom et al., 2004; Farina, 1999). Thus, in this paper, we use the LS estimator in (34) to calculate the estimate of $\hat{\boldsymbol{\chi}}$ for reducing the computational burden. As noted in Remark 12, the bias can be significantly reduced by appropriately choosing the parameter of k .

Once the target position is estimated by (34), for a given observation radius r and n vehicles, the optimal assignment procedure between the optimal deployment points and the vehicles can be realized by two steps. First, the n optimal deployment points can be calculated by

$$\begin{cases} x_j^* = \hat{x}_t - r \cos(\theta_j^*) \\ y_j^* = \hat{y}_t - r \sin(\theta_j^*) \end{cases} \quad (36)$$

where θ_j^* is obtained just by using the indices j instead of i in θ_i^* for avoiding confusion, and θ_i^* is given by (20), $j \in \{1, 2, \dots, n\}$. Then, some assignment strategy is adopted to allocate the optimal deployment points of $\mathbf{m}_j = [x_j^*, y_j^*]^T$ to the vehicles of $\mathbf{s}_i = [x_i, y_i]^T$, $i, j \in \{1, 2, \dots, n\}$.

Here, for improving the computational efficiency, we propose a 0–1 integer programming method to solve this assignment problem. First, the above-mentioned assignment problem is transformed into a 0–1 integer programming problem

$$\begin{aligned} \min \quad & \gamma = \sum_{i=1}^n \sum_{j=1}^n d_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i=1}^n x_{ij} = 1, \quad j \in \{1, 2, \dots, n\} \\ & \sum_{j=1}^n x_{ij} = 1, \quad i \in \{1, 2, \dots, n\} \\ & x_{ij} = 0 \text{ or } 1 \end{aligned} \quad (37)$$

where $d_{ij} = \|\mathbf{s}_i - \mathbf{m}_j\|$. Then, the 0–1 integer programming problem in (37) can be solved by resorting to Matlab© routine *linprog*. Finally, the assignment results between the optimal deployment points and the vehicles can be determined.

Now the optimal deployment algorithm of vehicles for multi-target localization is presented as follows.

Algorithm 1. Optimal deployment algorithm of vehicles for multi-target localization

- STEP 1 Set $h := 1$, $h_{\max} = 2$ (by default). Initialize $\chi^p, p \in \{1, 2, \dots, U\}$ and $\mathbf{s}_i, i \in \{1, 2, \dots, n\}$. Give the desired observation radius of r_p for each target.
- STEP 2 Solve (31) to determine the optimal allocating number n_p of vehicles for each target.
- STEP 3 Determine θ_1^p and k^p for each target according to n_p by (20).
- STEP 4 If $h = 1$, collect $z_i^p, i \in \{1, 2, \dots, n\}, p \in \{1, 2, \dots, U\}$; otherwise, collect $z_i^p, i \in \{1, 2, \dots, n_p\}$ for each assigned target. Calculate $\hat{\chi}^p$ by (34) and $\mathbf{m}_j^p$ by (36), respectively, $j \in \{1, 2, \dots, n_p\}$. Then collect $\mathbf{m}_j^p$ and rearrange them to $\mathbf{m}_j, j \in \{1, 2, \dots, n\}$.
- STEP 5 Solve (37) and then deploy the vehicle $\mathbf{s}_i$ to the optimal deployment point $\mathbf{m}_j$ by setting $\mathbf{s}_i = \mathbf{m}_j$.
- STEP 6 $h := h + 1$. If $h > h_{\max}$, exit; otherwise, goto STEP 4.

Once the deployment of vehicles is completed, the estimates of $\hat{\chi}^p = [\hat{x}_t^p, \hat{y}_t^p]^T$ for target p can be determined by the continuous observations of the assigned n_p vehicles. Meanwhile, we can further use the means of the estimates of $\hat{\chi}^p = [\hat{x}_t^p, \hat{y}_t^p]^T$ instead of $\hat{\chi}^p = [\hat{x}_t^p, \hat{y}_t^p]^T$ to obtain much higher estimation accuracy with the increase of observation times for continuously surveillance requirements. Moreover, by (36), one can observe that the optimal deployment errors of vehicles are equal to the location estimation errors of targets. Fortunately, however, the variances of target location estimation can attain the CRLB in (28) using any one of the proposed optimal sensor-target geometries with any unbiased estimator.

4. An illustrative example

In this section, an illustrative example is used to verify the proposed approach. For comparison, the MLE-based approach is also adopted, which is obtained just by using (35) instead of (34) in STEP 4 of Algorithm 1 for the calculation of the estimate $\hat{\chi}$. We denote the LS- and MLE-based optimal deployment algorithms as LS-DA and MLE-DA for simplicity, respectively. For examining the performance of these two algorithms, Monte Carlo simulations are performed. The means and root mean squared errors of the estimates $\hat{x}_t$ and $\hat{y}_t$ are respectively defined by

$$\bar{\hat{x}}_t = \frac{1}{M_c} \sum_{g=1}^{M_c} \hat{x}_t(g), \quad \sigma_{\hat{x}_t} = \sqrt{\frac{1}{M_c} \sum_{g=1}^{M_c} [x_t - \hat{x}_t(g)]^2} \quad (38)$$

$$\bar{\hat{y}}_t = \frac{1}{M_c} \sum_{g=1}^{M_c} \hat{y}_t(g), \quad \sigma_{\hat{y}_t} = \sqrt{\frac{1}{M_c} \sum_{g=1}^{M_c} [y_t - \hat{y}_t(g)]^2} \quad (39)$$

where M_c is the total times of Monte Carlo runs, and $\hat{x}_t(g)$ and $\hat{y}_t(g)$ denote the estimates of the true values x_t and y_t at the g th Monte Carlo run, respectively. Then we define

$$E_{\text{RMS}} = \sqrt{\sigma_{\hat{x}_t}^2 + \sigma_{\hat{y}_t}^2} \quad (40)$$

as a performance measure, which is used to verify the closeness of E_{RMS} obtained by the proposed algorithm to the true minimum of GDOP with circular formation.

Table 1

Simulation parameters for multi-target localization.

Parameter	Target 1	Target 2	Target 3
χ^p	$[600, 700]^T$	$[800, -500]^T$	$[-500, 0]^T$
r_p	400	500	600
n_p	2	3	3
θ_1^p	0	π	π
k^p	1	2	1

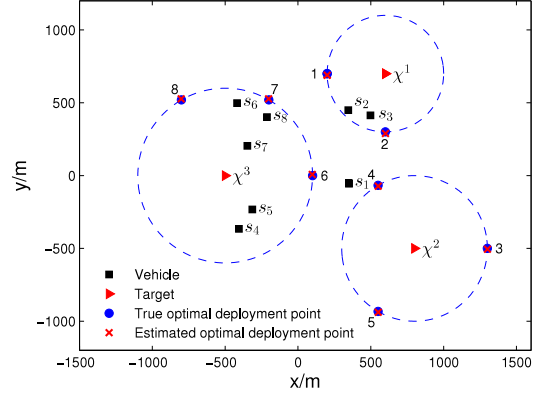


Fig. 5. The sensor-target geometry before the deployment of vehicles using LS-DA.

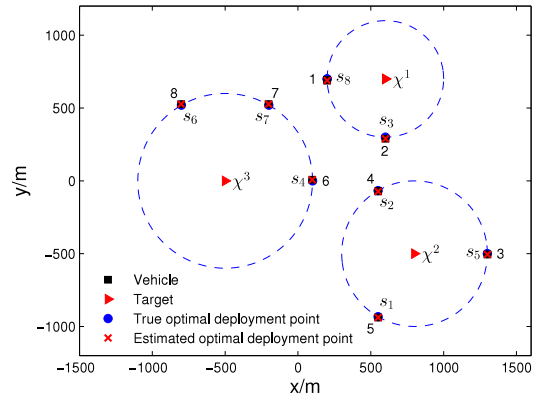


Fig. 6. The sensor-target geometry after the deployment of vehicles in a single run using LS-DA.

Consider the case of $n = 8$ vehicles used to localize the positions of $U = 3$ targets. For this case, the branch and bound method in Li and Sun (2006) is used in STEP 2 of Algorithm 1. Set $M_c = 5000$, $h_{\max} = 2$, $\sigma_\theta = 1^\circ$, and the remaining simulation parameters are listed in Table 1. By default, the length units are in meters. The vehicles are first randomly distributed around the targets, as shown in Fig. 5. Then the assignment results of the vehicles are obtained as $\mathbf{s}_1 \rightarrow \mathbf{m}_5$, $\mathbf{s}_2 \rightarrow \mathbf{m}_4$, $\mathbf{s}_3 \rightarrow \mathbf{m}_2$, $\mathbf{s}_4 \rightarrow \mathbf{m}_6$, $\mathbf{s}_5 \rightarrow \mathbf{m}_3$, $\mathbf{s}_6 \rightarrow \mathbf{m}_8$, $\mathbf{s}_7 \rightarrow \mathbf{m}_7$, $\mathbf{s}_8 \rightarrow \mathbf{m}_1$, shown in Fig. 6. The simulation results after the vehicles deployed are given in Table 2. It is observed that for each target, the values of E_{RMS} are all expected to approach the minima of GDOP for both LS-DA and MLE-DA, which coincides with the theoretical result of Theorem 8.

For the comparison of computing efficiency, the average execution times of LS-DA and MLE-DA are calculated as 0.3 ms and 17.5 ms, respectively. The key parameters of the Monte Carlo running environment are listed as below:

- Operating system: Windows 10 Pro 64;
- CPU: Intel Core i7-7700 HQ Processor;

Table 2

Simulation results of the optimal deployment of vehicles for multi-target localization.

Parameter	Target 1		Target 2		Target 3	
	LS-DA	MLE-DA	LS-DA	MLE-DA	LS-DA	MLE-DA
x_t	600	600	800	800	−500	−500
$\hat{x}_t$	599.98	600.19	800.00	800.01	−500.13	−499.85
$\sigma_{x_t}^{\text{CRLB}}$	6.98	6.98	7.13	7.13	8.55	8.55
σ_{x_t}	7.00	6.98	7.18	7.16	8.62	8.57
y_t	700	700	−500	−500	0	0
$\hat{y}_t$	699.99	699.85	−500.00	−499.98	0.00	0.20
$\sigma_{y_t}^{\text{CRLB}}$	6.98	6.98	7.13	7.13	8.55	8.55
σ_{y_t}	6.98	7.03	7.13	7.13	8.56	8.56
E_{RMS}	9.89	9.91	10.12	10.10	12.15	12.11
GDOP	9.87	9.87	10.08	10.08	12.09	12.09

- Memory: 8GB(4 + 4) DDR4 2400 MHz;
- Matlab: 8.2.0.701 (R2013b).

As expected, LS-DA has lower average execution time than MLE-DA. Thus, in connection with the target localization accuracy and the computing efficiency, one would prefer LS-DA for the optimal deployment of vehicles for bearings-only multi-target localization.

5. Conclusion

In this paper, the optimal deployment of n vehicles with circular formation for 2D bearings-only multi-target localization is considered. A general explicit formula of GDOP is first derived, which can be used to evaluate the localization accuracy with different sensor-target geometries. Based on this generalized formula of GDOP, the necessary and sufficient condition for sensor placement optimality with circular formation is presented. All the optimal deployment solutions of vehicles are given with closed form. Moreover, an efficient optimal deployment algorithm of vehicles is developed for multi-target localization. In future work, we will investigate how to control multiple vehicles to arrive at the desired positions and then extend the method for target tracking.

Appendix A. Proof of Lemma 4

Proof. We first prove the closed form of $f(x)$ in (10). If $x = k\pi$, $k \in \mathbb{Z}$, then $\sin(ix) = 0$, $i \in \{1, 2, \dots, n-1\}$, which yields $f(x) = 0$. If $x \in \mathbb{R}$, $x \neq k\pi$, $k \in \mathbb{Z}$, then $\sin(x) \neq 0$ and (9) can be expanded as

$$f(x) = \frac{1}{2} \left\{ \frac{n(n-1)}{2} - n \sum_{i=1}^{n-1} \cos(2ix) + \sum_{i=1}^{n-1} i \cos(2ix) \right\} \quad (\text{A.1})$$

Calculating the sum terms of (A.1) in closed form and then using the trigonometric function transformations, one obtains

$$f(x) = \frac{n^2 \sin^2(x) - \sin^2(nx)}{4 \sin^2(x)} \quad (\text{A.2})$$

which coincides with the closed form of $f(x)$ in (10).

For proving the closed form of $df(x)/dx$ in (11), first, taking the derivative with respect to x at the two sides of (9), we have

$$\frac{df(x)}{dx} = \sum_{i=1}^{n-1} i(n-i) \sin(2ix) \quad (\text{A.3})$$

If $x = k\pi$, $k \in \mathbb{Z}$, then $\sin(2ix) = 0$, $i \in \{1, 2, \dots, n-1\}$, which leads to $df(x)/dx = 0$. If $x \in \mathbb{R}$, $x \neq k\pi$, $k \in \mathbb{Z}$, then $\sin(x) \neq 0$,

taking the derivative versus x at the two sides of (A.2), after some tedious calculation, then we obtain

$$\frac{df(x)}{dx} = \frac{\sin(nx)[\sin(nx) \cos(x) - n \cos(nx) \sin(x)]}{2 \sin^3(x)} \quad (\text{A.4})$$

Thus, the closed form of $df(x)/dx$ in (11) holds.

Similarly, using (A.3), the second-order derivative of $f(x)$ is

$$\frac{d^2f(x)}{dx^2} = 2 \sum_{i=1}^{n-1} i^2(n-i) \cos(2ix) \quad (\text{A.5})$$

For $x = k\pi$, $k \in \mathbb{Z}$, $\cos(2ix) = 1$, $i \in \{1, 2, \dots, n-1\}$, using (A.5), we have

$$\frac{d^2f(x)}{dx^2} = 2 \sum_{i=1}^{n-1} i^2(n-i) = \frac{n^2(n^2-1)}{6} \quad (\text{A.6})$$

In a similar fashion, for the case of $x \in \mathbb{R}$, $x \neq k\pi$, $k \in \mathbb{Z}$, taking the derivative versus x at the two sides of (A.4), yields

$$\frac{d^2f(x)}{dx^2} = \frac{\beta(x)}{2 \sin^4(x)} \quad (\text{A.7})$$

where $\beta(x)$ is defined in (14). Thus the closed form of $d^2f(x)/dx^2$ of (12) is also true. Then this completes the proof. ■

Appendix B. Proof of Lemma 5

Proof. We will prove it by two steps.

(i) For $x = k\pi$, $k \in \mathbb{Z}$, using Lemma 4, $f(x) = 0$, $df(x)/dx = 0$ and $d^2f(x)/dx^2 = n^2(n^2-1)/6 > 0$ ($n \geq 2$), and from (9), for all $x \in \mathbb{R}$, it is observed that $f(x) \geq 0$, thus $x = k\pi$, $k \in \mathbb{Z}$ are the global minimum points of $f(x)$ with $f_{\min} = 0$.

(ii) For $x \in \mathbb{R}$, $x \neq k\pi$, $k \in \mathbb{Z}$, then $\sin(x) \neq 0$, using Lemma 4, we have

$$\frac{df(x)}{dx} = 0 \Leftrightarrow \alpha(x) = 0 \quad (\text{B.1})$$

(a) If $\exists \xi \in \mathbb{R}$, $\xi \neq k\pi$, $k \in \mathbb{Z}$, such that $\alpha(\xi) = 0$, i.e.

$$\sin(n\xi) \cos(\xi) - n \cos(n\xi) \sin(\xi) = 0 \quad (\text{B.2})$$

then from (B.2), it yields $df(x)/dx|_{x=\xi} = 0$, therefore ξ is the stationary point of $f(x)$. Submitting ξ into (12) and noting that $\xi \neq k\pi$, $k \in \mathbb{Z}$, we obtain

$$\frac{d^2f(x)}{dx^2} \Big|_{x=\xi} = \frac{(n^2-1) \sin^2(n\xi)}{2 \sin^2(\xi)} \quad (\text{B.3})$$

Now we show $\sin(n\xi) \neq 0$ by contradiction. Assume $\sin(n\xi) = 0$ is true, then it yields $\cos(n\xi) \sin(\xi) = 0$ from (B.2). Remembering $\xi \neq k\pi$, $k \in \mathbb{Z}$ and then $\sin(\xi) \neq 0$, which leads to $\cos(n\xi) = 0$, that contradicts the assumption of $\sin(n\xi) = 0$, thus $\sin(n\xi) \neq 0$. Then using (B.3), for $n \geq 2$, it follows that

$$\frac{d^2f(x)}{dx^2} \Big|_{x=\xi} > 0 \quad (\text{B.4})$$

Hence ξ is the local minimum point of $f(x)$. It follows immediately that all $\xi \in \mathbb{R}$, $\xi \neq k\pi$, $k \in \mathbb{Z}$ satisfying (B.2) are the local minimum points of $f(x)$.

(b) If $\sin(nx) = 0$, then the corresponding solutions are $x^* = k\pi/n$, $k \in \mathbb{Z}$. Note that $x^* \neq k\pi$, $k \in \mathbb{Z}$, submitting x^* into (12) then we have

$$\frac{d^2f(x)}{dx^2} \Big|_{x=x^*} = -\frac{n^2}{2 \sin^2(x^*)} < 0 \quad (\text{B.5})$$

Define the set of k as $\Phi = \{k \mid k \neq mn, k \in \mathbb{Z}, m \in \mathbb{Z}\}$, and then for all $x^* = k\pi/n$, $k \in \Phi$, the corresponding extreme values of

$f(x)$ are all equal to $f_{\max} = n^2/4$, thus $x^* = k\pi/n$, $k \in \Phi$ are the global maximum points of $f(x)$.

In combination with the analyses of steps (i) and (ii), it follows immediately that

$$x^* = k\pi/n \Leftrightarrow f_{\max} = n^2/4 \quad (\text{B.6})$$

where $k \in \Phi$, $\Phi = \{k \mid k \neq mn, k \in \mathbb{Z}, m \in \mathbb{Z}\}$. Thus the proof is completed. ■

Appendix C. Proof of Theorem 6

Proof. Denote $\mathbf{z} = \{z_i\}_{i=1}^n$, then the likelihood function of calculating the unknown χ can be described by

$$p(\mathbf{z}|\chi) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma_{\theta_i}} \exp\left\{-\frac{1}{2\sigma_{\theta_i}^2}[z_i - h_i(\chi)]^2\right\} \quad (\text{C.1})$$

and the corresponding FIM is defined as

$$\mathbf{J} = \mathcal{E} \left\{ [\nabla_{\chi} \ln p(\mathbf{z}|\chi)] [\nabla_{\chi} \ln p(\mathbf{z}|\chi)]^T \right\} \quad (\text{C.2})$$

Note that $\nabla_{\chi} \ln p(\mathbf{z}|\chi) = \sum_{i=1}^n \frac{1}{\sigma_{\theta_i}^2} [z_i - h_i(\chi)] \nabla_{\chi} h_i(\chi)$ and

$$\nabla_{\chi} h_i(\chi) = \begin{bmatrix} -\frac{y_t - y_i}{(x_t - x_i)^2 + (y_t - y_i)^2} \\ \frac{x_t - x_i}{(x_t - x_i)^2 + (y_t - y_i)^2} \end{bmatrix} = \begin{bmatrix} -\frac{\sin(\theta_i)}{d_i} \\ \frac{\cos(\theta_i)}{d_i} \end{bmatrix} \quad (\text{C.3})$$

After some calculation, we determine the FIM in (C.2) as

$$\mathbf{J} = \sum_{i=1}^n c_i^2 \begin{bmatrix} \sin^2(\theta_i) & -\sin(\theta_i)\cos(\theta_i) \\ -\sin(\theta_i)\cos(\theta_i) & \cos^2(\theta_i) \end{bmatrix} \quad (\text{C.4})$$

where $c_i = 1/(d_i\sigma_{\theta_i})$. If the inverse of $\mathbf{J}$ exists, then the CRLB can be written as

$$\text{CRLB} = \mathbf{J}^{-1} = \frac{\mathbf{J}^*}{\det(\mathbf{J})} \quad (\text{C.5})$$

with

$$\mathbf{J}^* = \sum_{i=1}^n c_i^2 \begin{bmatrix} \cos^2(\theta_i) & \sin(\theta_i)\cos(\theta_i) \\ \sin(\theta_i)\cos(\theta_i) & \sin^2(\theta_i) \end{bmatrix} \quad (\text{C.6})$$

and

$$\det(\mathbf{J}) = \sum_{1 \leq i < j \leq n} c_i^2 c_j^2 \sin^2(\theta_{ij}) \quad (\text{C.7})$$

where $\theta_{ij} = \theta_j - \theta_i$.

Finally, using (C.7) and (C.6) in (C.5), and remembering the definition of GDOP in (4), the GDOP is calculated as (17). The proof is thus completed. ■

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